

CSE 4345: Big Data Analytics

Week 2, Part 2 — Research Article & Case Study

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Today's Agenda

- 1 Research Paper — Halevy, Rajaraman & Ordille (2006)
- 2 Case Study — HL7 FHIR: Healthcare Data Integration at Scale
- 3 Live Exercise

Research Article Covered

- 1 Halevy, A., Rajaraman, A., & Ordille, J. (2006). *Data Integration: The Teenage Years*. VLDB 2006. Seoul, Korea.

Case Study

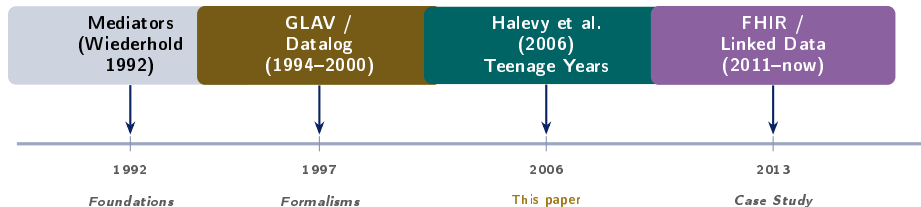
HL7 FHIR in US Healthcare — national-scale federated health data integration under the 21st Century Cures Act (2016).

CLO Addressed

CLO2 — Analyse real-world data integration challenges
PLO(b,c)

Research Paper — Halevy, Rajaraman & Ordille (2006)

Research Landscape: Data Integration Timeline



Why This Sequence?

Wiederhold (1992) introduced mediator architectures → the GLAV decade formalised schema mappings → Halevy et al. (2006) honestly audited what *hadn't* worked and proposed the **pay-as-you-go** paradigm → FHIR / Linked Data put those lessons into production. Week 2 builds on all four layers.

Halevy et al. (2006): Publication Context

Full Citation

Halevy, A., Rajaraman, A., & Ordille, J. (2006). **Data Integration: The Teenage Years**. In *Proceedings of the 32nd International Conference on Very Large Data Bases (VLDB)*, pp. 9–16. Seoul, Korea.

Venue: VLDB — Rank A* (CORE), top-tier DB venue

Cited by: >1,600 (Google Scholar, 2024)

Authors: Alon Halevy (Google), Anand Rajaraman (Amazon A9), Jayant Ordille (Yahoo! Research)

Historical Moment

By 2006, data integration research was ≈ 15 years old (“teenage”). Three industry heavyweights wrote an honest retrospective — celebrating progress, naming failures, and redirecting the field.

The Core Question

“Why, after 15 years of research, do enterprises still struggle to integrate their data?”

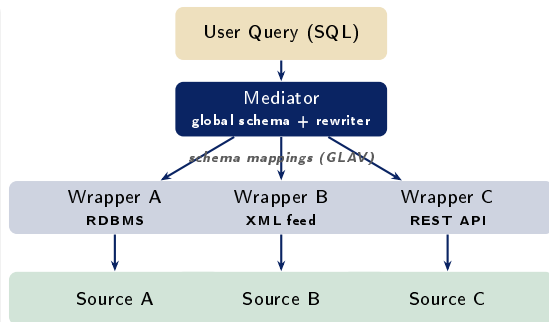
The Answer (in one sentence)

Because perfect schema mappings require **semantic understanding** that cannot be automated — so the field must shift from *completeness* to *incremental, probabilistic integration*.

Technical Contribution 1 — The Integration Stack

The Classic Mediator–Wrapper Architecture

- **Wrappers** expose each data source through a uniform query interface (relational, XML, or object)
- **Mediator** holds a **global schema** and rewrites queries into sub-queries against wrappers
- **Schema mappings** (the hard part): declarative rules that translate between source and global schemas
- The dominant formalism: **GLAV** (Global-Local-As-View) — sources defined as views over the global schema *and* vice versa; allows bidirectional query rewriting



The Semantic Gap Problem

Even with perfect GLAV mappings, the *meaning* of `customer_id` in the Sales DB $\neq$ `cust_no` in the CRM — no formal system can resolve this without **domain expertise**.

Technical Contribution 2 — Pay-As-You-Go Integration

The Paper's Central Proposal

Traditional integration demands **complete, correct schema mappings before any query can run**. This is the primary reason real projects fail or are abandoned after months of schema work.

Pay-as-you-go (PAYG) integration (Halevy et al. 2006):

- 1 Start with *imperfect* mappings (even keyword-level)
- 2 Return *ranked, uncertain* answers rather than blocking on completeness
- 3 Improve mappings *incrementally* as usage reveals gaps
- 4 Track **data provenance** so downstream consumers know source quality

Why Schema Matching Fails at Scale

- n schemas $\Rightarrow O(n^2)$ pairwise mappings needed
- Each mapping requires **semantic consensus** from domain owners (weeks of meetings)
- Sources evolve; mappings **rot silently**
- No enterprise has ever fully mapped >50 source schemas using the classical approach

Dataspace Systems (Halevy et al. PODS 2006)

A **dataspace** manages heterogeneous data

Technical Contribution 3 — Semantic Heterogeneity Taxonomy

Four Levels of Mismatch (Halevy et al. 2006)

Level	Example
Naming	cust_id vs. customer_no
Structural	flat table vs. nested JSON
Semantic	revenue (gross) vs. revenue (net)
Temporal	daily snapshot vs. event stream

Key Insight

Naming and structural mismatches can be solved by schema matching algorithms (ML-based, 2024).

Halevy's Prescription

- **Invest in metadata:** every dataset should carry a machine-readable description of semantics (today: data catalogs — Apache Atlas, DataHub, Alation)
- **Embrace uncertainty:** integration results should carry **confidence scores and provenance lineage** (today: dbt lineage, OpenLineage)
- **Domain ownership:** the producing team is the only reliable author of semantic documentation (today: **Data Mesh** principle 1)
- **Probabilistic mapping:** use Bayesian schema matching rather than deterministic rules

Concepts from the Paper Still in Active Use

- **Data catalogs** — Apache Atlas, DataHub, Alation implement “dataspace co-existence” as their core model
- **Schema registry** — Confluent Schema Registry, AWS Glue Catalog apply PAYG registration
- **Data lineage** — OpenLineage, dbt implements provenance tracking proposed in the paper
- **Data Mesh** — Dehghani (2019) explicitly cites Halevy’s “domain ownership” prescription as an ancestor
- **LLM-based schema matching** — GPT-4 / Gemini used as mediators; Halevy himself co-authored work at Google on this (2023)

Stanford CS246 / CMU 15-721 Perspective

- Stanford CS246 covers schema matching as a **set-similarity / embedding** problem (Jaccard + word2vec on column names)
- CMU 15-721 frames integration as a **query federation** problem: how to push predicates through heterogeneous wrappers efficiently
- Both courses trace the theoretical lineage directly to Halevy et al. 2006

CLO2 Connection

By analysing *why* classical integration fails, students can design systems that avoid the

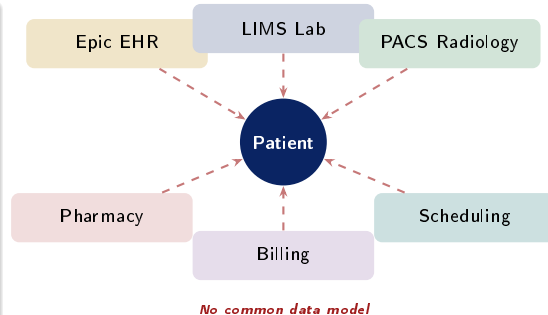
$O(n^2)$ schema-mapping trap — the **core CLO2**

Case Study — HL7 FHIR: Healthcare Data Integration at Scale

The Healthcare Data Integration Problem

Why Healthcare Data is the Hardest Integration Domain

- A typical US hospital has **50–400 distinct systems**: EHR (Epic, Cerner), lab (LIMS), radiology (PACS), pharmacy (Pyxis), billing (IDX), scheduling, and dozens more
- Each system uses **its own data model and vocabulary**
- Patient identity matching across systems: no universal patient ID in the US pre-2016
- **HIPAA** (1996) requires strict access control; integration cannot expose PHI
- Result: clinicians make decisions with **incomplete data**; studies estimate 400,000 preventable deaths/year in the US from medical



HL7 FHIR: The Standard That Changed Healthcare Integration

What is HL7 FHIR?

Health Level Seven (HL7) International is a standards body founded in 1987. **FHIR** (Fast Healthcare Interoperability Resources), released as Release 1 in 2014 and R4 in 2019, defines:

- **Resources:** atomic clinical data objects (Patient, Observation, Medication, Encounter — 145+ types)
- **REST API:** GET /Patient/{id}, POST /Observation, standard HTTP verbs
- **SMART on FHIR:** OAuth2-based authorisation for third-party app access
- **Terminology binding:** SNOMED CT, LOINC, ICD-10 as controlled vocabularies (solving semantic heterogeneity!)

Key FHIR Resource (JSON)

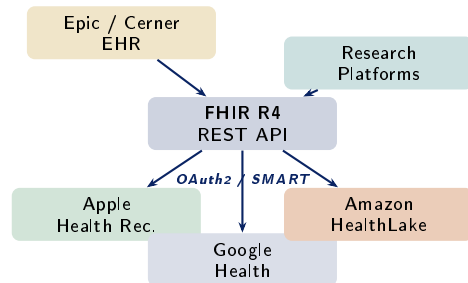
```
{
  "resourceType": "Observation",
  "status": "final",
  "code": {
    "coding": [{
      "system": "http://loinc.org",
      "code": "8480-6",
      "display": "Systolic BP"
    }]
  },
  "valueQuantity": {
    "value": 128,
    "unit": "mmHg"
  }
}
```

Case Study: 21st Century Cures Act — National FHIR Mandate

The Mandate (US, 2016 / enforced 2021)

The **21st Century Cures Act** required every certified EHR vendor in the US to expose patient data via a **FHIR R4 API by April 2021**. Penalties for *information blocking*: up to \$1M per violation.

- **Scale:** >350M patients, >6,000 hospitals, >900,000 clinicians affected
- **Impact:** Apple Health Records, Google Health, Amazon HealthLake — all built on this mandate
- Every major EHR (Epic, Cerner/Oracle, Allscripts) deployed FHIR endpoints in 2020–2021



Integration Lesson

Regulatory mandate succeeded where 30 years of

ETL Architecture Inside a FHIR-Based Health Data Lake

Typical Hospital FHIR ETL Pipeline

- 1 **Extract:** Nightly or real-time FHIR API pull (GET /Patient?\$everything) — returns a Bundle of all resources for a patient
- 2 **Transform:** FHIR → flat Parquet tables via Apache Spark + `fhir.resources` Python library; explode nested arrays, resolve LOINC/SNOMED codes
- 3 **Load:** Delta Lake / Apache Iceberg on S3; partitioned by `encounter_date` and `patient_id`
- 4 **Governance:** column-level masking for PHI (Azure Purview / AWS Macie), RBAC per department
- 5 **Analytics:** downstream BI (PowerBI), cohort

Scale Numbers — Mayo Clinic FHIR Platform

- **Source systems:** 35 EHR instances merged
- **FHIR resources:** 4.2 billion stored resources
- **Patients:** 1.3 million active patient records
- **Daily ingestion:** $\approx 12\text{M}$ new FHIR resources/day
- **Query latency:** < 2 seconds on 5-year patient history (via columnar Parquet + predicate pushdown)
- **Schema convergence:** from 35 distinct schemas to **one FHIR R4 global schema** — solving the Halevy $O(n^2)$ mapping problem through *regulatory standardisation*

Lessons Learned & Discussion Questions

Key Lessons from the FHIR Case

- 1 **Canonical data model wins:** all 35 EHR schemas converged to FHIR R4 — the “global schema” Halevy’s mediator model required but couldn’t enforce
- 2 **Standards adoption needs incentives:** HIPAA (1996) and CDA failed; the \$1M penalty worked where interoperability appeals had not
- 3 **Semantic layer is still hard:** LOINC codes for the *same* lab test differ by lab vendor; a mapping table of >200,000 entries is still maintained manually
- 4 **Provenance is non-negotiable:** every Observation in the lake carries the FHIR source URL, extraction timestamp, and transformer version

Discussion Questions (CLO2)

- 1 Halevy et al. argue that **semantic heterogeneity** cannot be solved algorithmically. FHIR uses **LOINC as a controlled vocabulary**. Does LOINC fully solve semantic heterogeneity, or does it only defer it? Justify with an example. *[CLO2 — Analyse]*
- 2 The 21st Century Cures Act used **regulatory mandate** to force FHIR adoption. Propose an analogous policy intervention for **financial services data integration** (banks, insurers, payment processors). *[CLO2 — Evaluate]*
- 3 A hospital’s FHIR ETL ingests 12M resources/day. The Spark transformation

Live Exercise

Live Exercise — FHIR API Exploration

Task (20 minutes, pairs)

The HAPI FHIR public test server exposes a live FHIR R4 API.

Step 1 — Explore patient resources:

```
curl https://hapi.fhir.org/baseR4/Patient?_count=5
```

Step 2 — Fetch observations for a patient:

```
curl https://hapi.fhir.org/baseR4/
```

```
Observation?patient={id}&_format=json
```

Step 3 — Identify the integration challenges:

- Are LOINC codes present on every Observation?
- What field carries patient identity?
- How would you flatten the `Bundle.entry[]` array into a Spark DataFrame?

Python Skeleton (PySpark)

```
import requests, json
from pyspark.sql import SparkSession

url = ("https://hapi.fhir.org"
      "/baseR4/Observation"
      "?_count=50&_format=json")
bundle = requests.get(url).json()
entries = bundle.get("entry", [])

rows = []
for e in entries:
    r = e["resource"]
    rows.append({
        "id": r.get("id"),
        "status": r.get("status"),
        "loinc": (r.get("code", {})
                  .get("coding", [{}])[0]
                  .get("code", "?")),
        "value": (r.get(
            "valueQuantity", {}
            ).get("value"))
    })
```

References

Research Article

- Halevy, A., Rajaraman, A., & Ordille, J. (2006). Data integration: The teenage years. *Proceedings of the 32nd VLDB*, 9–16.

Textbooks [SA], [TE], [TW]

- **[SA]** Acharya, S., & Chellappan, S. (2015). *Big Data and Analytics* (Ch. 2 — Data Integration). Wiley.
- **[TE]** Erl, T., Khattak, W., & Buhler, P. (2016). *Big Data Fundamentals* (Ch. 4 — ETL/ELT Architecture). Prentice Hall.
- **[TW]** White, T. (2015). *Hadoop: The Definitive Guide*, 4th ed. (Ch. 3 — HDFS for integration pipelines). O'Reilly.

Case Study Sources

- HL7 International. (2019). *FHIR R4 Specification*. hl7.org/fhir/R4
- U.S. Department of Health & Human Services. (2020). *21st Century Cures Act: Interoperability*.

Key Takeaways

- Halevy et al. (2006) proved that **semantic heterogeneity cannot be solved by algorithms alone** — domain knowledge and incremental PAYG integration are the realistic path
- FHIR is the most successful real-world deployment of the **global schema / mediator architecture** — but it required a **regulatory mandate** to achieve adoption
- **Data provenance** and **controlled vocabularies** (LOINC, SNOMED) are the operational counterparts of Halevy's theoretical prescriptions
- The lessons apply equally to finance, logistics, and any domain with many heterogeneous source systems

Quote of the Week

“Data integration is the most important problem in information systems today, and arguably the least glamorous.”

— Halevy, Rajaraman & Ordille (2006)

Part 1 Preview — Week 3

HDFS Architecture & the Hadoop Ecosystem

- NameNode / DataNode replication model
- Rack-aware block placement
- Hadoop ecosystem: Hive, Pig, HBase